

Patrick Norrick

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Research Interests

Dynamic Macroeconomics; Demographics & Population Dynamics; Economic Growth; Computational Methods; Time Series Econometrics

Education

Washington University in St. Louis

B.A. Mathematics and Economics

Minor in Computer Science

GPA: 3.97/4.0 (4.0 in majors)

Graduated: May 2022

Selected Coursework

Mathematics: Numerical Optimization (graduate-level), Real Analysis, Measure Theory, Linear Algebra, Differential Equations, Probability Theory, Mathematical Statistics

Econometrics & Quantitative Methods: Applied Econometrics, Time Series Analysis, Computational Macroeconomics, Stochastic Processes

Economics: Game Theory, Asset Pricing

Research Experience

American Enterprise Institute (AEI) (Sept. 2023 - Present)

Washington, D.C.

Research Associate

- **Coauthoring** paper on demographic projections and long-run growth with Dr. Jesús Fernández-Villaverde. Preliminary results discussed in his invited presentations at the NBER Annual Conference on Macroeconomics (Apr. 2025), CEPR–EBRD Research Symposium (Jun. 2025), and Penn Population Studies Colloquium (Sept. 2024).
- Analysis of Bayesian simulation models and Markov Chain Monte Carlo applied to demographic projections in Python and R.
- Coauthored AEI working papers on demographic economics, labor market dynamics, and human capital (with Dr. Nicholas Eberstadt).

NISA Investment Advisors, LLC (Jun. 2022 – Aug. 2023)

St. Louis, MO

Quantitative Research Analyst

- Developed and enhanced macro-finance models in MATLAB and Python for portfolio management and simulation of term-structure dynamics.
- Extended proprietary stochastic interest rate model (originating with Nobel Laureate Philip Dybvig) for macroeconomic analysis.

WashU Dept. of Economics (May 2020 – Jun. 2021)

St. Louis, MO

Undergraduate Research Assistant to Dr. Ian Fillmore

- Conducted microdata analysis of employment effects of minimum wage increases.
- Contributed empirical analysis and draft text to “*A \$15 Federal Minimum Wage Is Outside Historical Experience*” (Research in Economics, 2022). Acknowledged for contributions.
- Performed literature reviews on intergenerational mobility and consumer choice.

Working Papers & Publications

- **With Jesús Fernández-Villaverde:** *Demographic Projections and Long-Run Growth: A Review of the UN Population Division’s World Population Prospects*. Draft in progress, targeted for NBER Working Paper Series submission (2025).
- With Nicholas Eberstadt: *America’s Immigration Mess: An Illustrated Guide* (AEI Working Paper, 2025).
- With Nicholas Eberstadt, Radek Sabatka, & Peter Van Ness: *Knowledge and Skills in China’s K-12 Population: An Inquiry into Knowledge Capital in the PRC*. (AEI Working Paper, 2024). Book-length report; Received feedback from Eric Hanushek & Scott Rozelle.

Honors & Awards

- NISA Scholars Award in Economics, Washington University in St. Louis, 2022
- Distinction in Mathematics, 2022
- College Honors in Arts & Sciences, 2022
- Omicron Delta Epsilon (Economics Honor Society), 2022
- Lisa and Alan Engel Scholarship (School of Arts & Sciences), 2021
- Dean’s List, all semesters
- US Track & Field and Cross Country All-Academic, 2019 & 2021

Teaching & Mentorship

- **NISA Scholars Program** (Spring 2023): Gave lectures on real analysis and measure theory for colleagues seeking to strengthen mathematical foundations and engage with research in stochastic calculus. Topics included basic topology, sequences, continuity and convergence, the Riemann-Stieltjes integral, sequences and series of functions, measurability, Lebesgue integration, and basic functional analysis.
- **American Enterprise Institute** (2024-2025): Supervised and mentored four undergraduate and graduate research interns in economics and data analysis.

Technical Skills

Programming: Julia, Python, MATLAB, L^AT_EX, R, Stata, C++

Tools: Git/GitHub, Unix, Jupyter, SQL, Java

Additional

NCAA Division III Varsity Cross Country and Track & Field (4-year letterwinner)